

	AMPACITY	MVA RATING	COND TEMP	SHEATH TEMP	TOTAL LOSS	STATUS
BURIED	821 A	187.81 MVA	90.0 °C	78.7 °C	34.924 W/m	Converged
FREE AIR	1552 A	354.89 MVA	90.0 °C	49.9 °C	126.244 W/m	Converged

1. Design Data & Input Parameters

Parameter	Formula / Symbol	Value	Unit
Conductor material		Cu	
Conductor size		630	mm2
Conductor type		stranded_compact	
Conductor diameter Dc		30.3	mm
Conductor screen OD Dc_sc		33.3	mm
Max conductor temp theta_max		90	degC
Insulation material		XLPE	
Insulation outer diameter Di		64.3	mm
Insulation screen OD Di_sc		66.9	mm
Insulation thickness t_ins		15.50	mm
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.001000	
Screen type		Al	
Screen mean diameter ds		67.7	mm
Screen area As		170.2	mm2
Armour type		none	
Oversheath material		PE	
Cable outer diameter De		75.5	mm
Rated voltage U		132.0	kV
U0 line-to-earth		76.210	kV
Frequency		50	Hz
Bonding		solid	
Formation		trefoil	
Depth to cable centre L		1000	mm
Axial spacing s		75.5	mm
Ambient ground temp theta_a		20	degC
Soil thermal resistivity rho_soil		1.00	K.m/W
Ambient air temp theta_a_air		25	degC
Air mounting		free	
Solar radiation		H=1000 W/m², σ=0.40	
Wind speed		None	

2. DC Resistance [IEC 60287-1-1 Cl.2.1.1]

Parameter	Formula / Symbol	Value	Unit
IEC 60228 R0 at 20 degC		2.8300e-05	ohm/m
Temperature coefficient a20		3.9300e-03	1/K
R0 at theta_amb = 20 degC		2.8300e-05	ohm/m
R0 at theta_max = 90 degC		3.6085e-05	ohm/m

3. AC Resistance at theta_max [IEC 60287-1-1 Cl.2.1.2, 2.1.4]

Parameter	Formula / Symbol	Value	Unit
ks strand constant (IEC 60287-1-1 Table 2)		1.0000	
kp proximity constant (IEC 60287-1-1 Table 2)		1.0000	
$xs^2 = ks \cdot (8\pi f / R0) \cdot 1e-7$		3.4824042386	
$xs_{\text{eff}} = (xs^2)^2$		12.1271392811	
ys skin effect factor		0.0601241268	
$xp_{\text{eff}} = (xp^2)^2$		12.1271392811	
yp" intermediate proximity		0.0601241268	
yp proximity factor		0.0351000648	
R at theta_max AC = $R0(1+ys+yp)$		3.9521526380e-05	ohm/m

4. Dielectric Losses [IEC 60287-1-1 Cl.2.2]

Parameter	Formula / Symbol	Value	Unit
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.001000	
U0 line-to-earth		76.2100	kV
Di insulation outer diameter		64.3000	mm
Dc_sc conductor screen OD		33.3000	mm
Capacitance C = $er / (18 \ln(Di/Dc_sc)) \cdot 10^{-9}$		0.211077	nF/m
$\omega = 2 \pi f$		314.159265	rad/s
$Wd = \omega C U0^2 \tan(\delta)$		3.8513583657e-01	W/m
TB880 GP7		Applied	

5. Screen / Sheath Setup [IEC 60287-1-1 Cl.2.3]

Parameter	Formula / Symbol	Value	Unit
Screen material		Al	
Screen resistivity rho_s		2.8264e-08	ohm.m
Screen area As		170.1500	mm2
Wire/sheath outer OD D_sc_outer		68.5000	mm
Lay factor LFs		1.000000	
Screen Rs0 at 20 degC (with LFs)		1.6611e-04	ohm/m

6. Armour Losses [IEC 60287-1-1 Cl.2.4]

Parameter	Formula / Symbol	Value	Unit
Armour type		None	
lam2		0.000000	

7. Thermal Resistances [IEC 60287-2-1]

Parameter	Formula / Symbol	Value	Unit
rho_ins insulation		3.50	K.m/W
rho_sc semiconducting screens		2.50	K.m/W
rho_tape bedding tape under wire screen		6.00	K.m/W
T1_cs conductor screen		0.0375644199	K.m/W
T1_i insulation		0.3665350785	K.m/W
T1_si insulation screen		0.0157719906	K.m/W
T1_uwbt bedding tape under wire screen		0.0000000000	K.m/W
T1_raw = sum of all layers		0.4198714890	K.m/W
T1 correction x1.00 (foil=False, cover=100.0%)		0.4198714890	K.m/W
T2 buried (screen to foil tape)		0.0000000000	K.m/W
T3 reference diameter D_start		68.5000	mm
T3_raw oversheath (single-layer)		0.0541996092	K.m/W
T3 buried x1.6 (touching trefoil factor)		0.0867193748	K.m/W
T3 air x1.0 (no touching factor in air)		0.0541996092	K.m/W
u = 2L/De (buried)		26.490066	
T4 buried (soil)		1.5946928925	K.m/W
T4 air (iterative — see air section)		see air section	

8. Electrical Parameters

Parameter	Formula / Symbol	Value	Unit
Inductance L		0.160439	uH/m
Reactance X = omega L		50.4033	uOhm/m
Pos-seq resistance R1		51.1864	uOhm/m
Pos-seq impedance Z1		71.8369	uOhm/m
Capacitance C		0.211077	nF/m
Charging current Ic		5.0536	mA/m

9. Sheath Losses at Convergence (Buried)

Parameter	Formula / Symbol	Value	Unit
theta_s at convergence		78.722027	degC
Rs at theta_s		2.0542e-04	ohm/m
X sheath reactance		5.0403e-05	ohm/m
lam1' circulating current		0.2951527738	
lam1" eddy current		0.0000000000	
lam1 = lam1' + lam1"		0.2951527738	
lam2 armour		0.000000	

10. Ampacity — Direct Buried [IEC 60287-1-1 Cl.1.4.1.1]

Parameter	Formula / Symbol	Value	Unit
Delta_theta = theta_max - theta_amb		70.00	K
Etop numerator		69.2715741012	
Ebot denominator		1.0266e-04	
I = sqrt(Etop/Ebot) exact		821.4442792644	A
I rounded TB880 GP2		821	A
MVA = sqrt(3) x U x I		187.8074	MVA
Iterations		5	
Convergence		YES	

11. Temperature Profile — Buried

Parameter	Formula / Symbol	Value	Unit
theta_c conductor target		90.0	degC
theta_c back-calculated		90.000000	degC
theta_s sheath surface		78.722027	degC
theta_j cable outer surface		75.693419	degC
theta_a ambient ground		20.0	degC

12. Loss Budget — Buried

Parameter	Formula / Symbol	Value	Unit
Conductor Wc = I ² x R		26.667968	W/m
Sheath Ws = lam1 x Wc		7.871125	W/m
Armour Wa = lam2 x Wc		0.000000	W/m
Dielectric Wd		3.851358e-01	W/m
Total		34.924229	W/m

13. Free Air Parameters [IEC 60287-2-1 Cl.2.2.1]

Parameter	Formula / Symbol	Value	Unit
Formation		trefoil	
Mounting		free	
Z constant (IEC Table 2)		0.9600	
E constant (IEC Table 2)		1.2500	
g exponent (IEC Table 2)		0.2000	
De* cable outer diameter		75.50	mm
De* in metres		0.075500	m
^g	De*	0.596471	
h = Z/(De*) ^g + E		2.8594661337	W/(m ² K ^{5/4})
Solar radiation		Yes	
Solar intensity H		1000.0	W/m ²
Solar absorption sigma		0.40	
Ambient air temperature		25.0	degC
Delta_theta = theta_max - theta_a_air		65.00	K

14. Free Air T4 Calculation [IEC 60287-2-1 Cl.2.2.1]

Parameter	Formula / Symbol	Value	Unit
$KA = \pi \cdot De \cdot h \cdot (T1 / (n(1 + \lambda m1)) + T3)$		0.2522151779	$K^{(-1/4)}$
$\Delta \theta d = Wd \cdot (1 / (1 + \lambda m1) - 0.5) \cdot T1$		0.0414919101	K
$\Delta \theta ds = \sigma \cdot De \cdot H \cdot (T1 / (n(1 + \lambda m1)) + T3)$		11.2304298528	K
$rhs = D\theta_{theta} - D\theta_{theta_d} - D\theta_{theta_ds}$		53.7280782371	K
$x = (D\theta_{theta_s})^{(1/4)}$ converged		12.7469516240	$K^{(1/4)}$
$D\theta_{theta_s} = x^4 = \theta_{theta_j} - \theta_{theta_a}$		26401.3023353640	K
$T4 = 1 / (\pi \cdot De \cdot h \cdot x)$		0.1156676416	K.m/W
Solar term $\sigma \cdot De \cdot H \cdot T4$		3.4931627771	K

15. Sheath Losses at Convergence (Air)

Parameter	Formula / Symbol	Value	Unit
θ_{theta_s} at convergence		49.937818	degC
R_s at θ_{theta_s}		1.8615e-04	ohm/m
X sheath reactance		5.0403e-05	ohm/m
$\lambda m1'$ circulating current		0.3217266493	
$\lambda m1''$ eddy current		0.0000000000	
$\lambda m1 = \lambda m1' + \lambda m1''$		0.3217266493	
$\lambda m2$ armour		0.000000	

16. Ampacity — Free Air [IEC 60287-1-1 Cl.1.4.4.1]

Parameter	Formula / Symbol	Value	Unit
Etop numerator (incl solar)		61.3605614785	
Ebot denominator		2.5467e-05	
$I = \sqrt{Etop/Ebot}$ exact		1552.2210172574	A
I rounded TB880 GP2		1552	A
$MVA = \sqrt{3} \times U \times I$		354.8854	MVA
Iterations		5	
Convergence		YES	

17. Temperature Profile — Free Air

Parameter	Formula / Symbol	Value	Unit
θ_{theta_c} conductor target		90.0	degC
θ_{theta_c} back-calculated		90.000000	degC
θ_{theta_s} sheath surface		49.937818	degC
θ_{theta_j} cable outer surface		43.095464	degC
θ_{theta_a} ambient air		25.0	degC

18. Loss Budget — Free Air

Parameter	Formula / Symbol	Value	Unit
Conductor $Wc = I^2 \times R$		95.222774	W/m
Sheath $Ws = \lambda m1 \times Wc$		30.635704	W/m
Armour $Wa = \lambda m2 \times Wc$		0.000000	W/m
Dielectric Wd		3.851358e-01	W/m
Solar heat $\sigma \cdot De \cdot H \cdot T4$		3.493163	W/m
Total		126.243614	W/m

19. Electrical Parameters

Parameter	Formula / Symbol	Value	Unit
Inductance L		0.160439	uH/m
Reactance X = omega L		50.4033	uOhm/m
Pos-seq resistance R1		51.1864	uOhm/m
Pos-seq impedance Z1		71.8369	uOhm/m
Capacitance C		0.211077	nF/m
Charging current Ic		5.0536	mA/m



Standards: IEC 60287-1-1:2023 | IEC 60287-2-1:2015 | CIGRE TB 880 guidance applied.
GP2: ampacity rounded down. GP6: eddy losses always calculated. GP7: dielectric losses always calculated. GP8: exact T4 formula. T3: 1.6x factor for touching trefoil (buried only).
Solar: H=1000 W/m², σ=0.40.
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